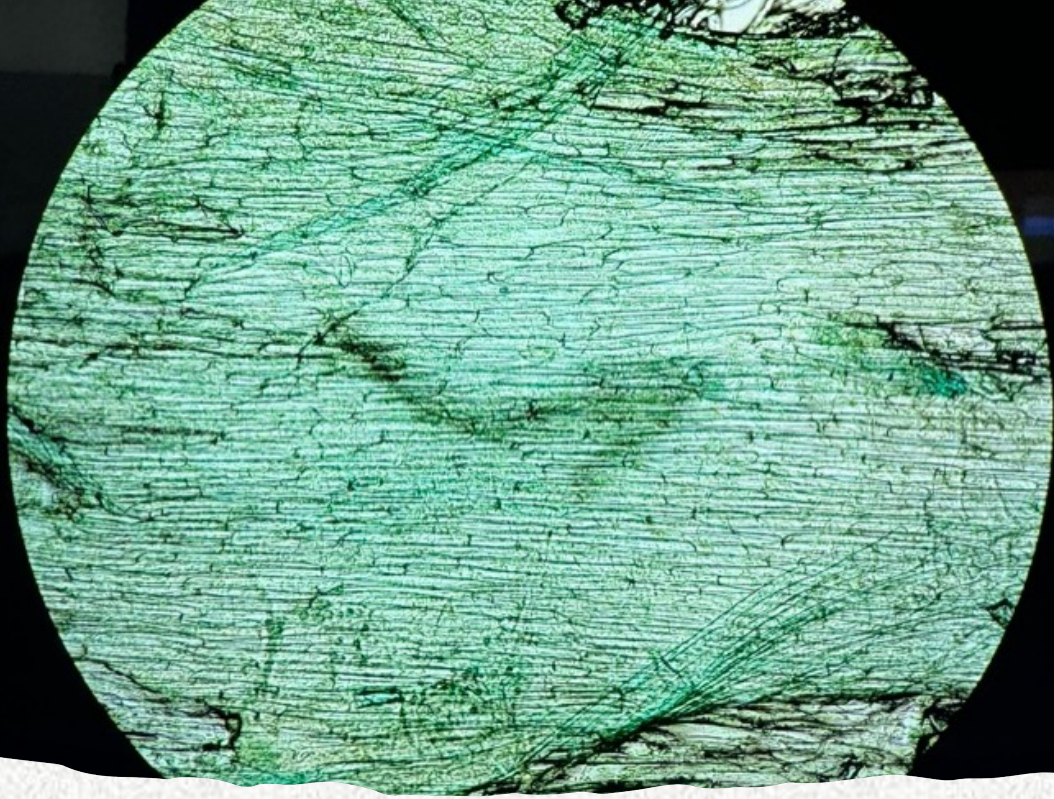
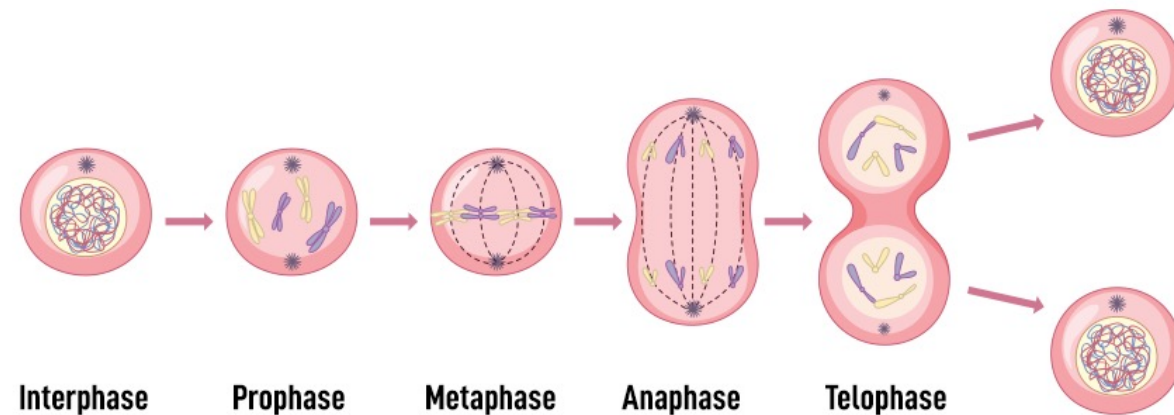




Mitosis in Onion Root Cells

Introduction

- Mitosis is a complex process in which cells divide to produce 2 new daughter cells. During mitosis the DNA of the cell's nucleus is split into 2 equal sets of chromosomes.
- All phases of mitosis are carefully regulated so that information is passed on correctly to the new cells and to limit cell division to when it is needed only. If that doesn't happen cell division can lead to cancer.



Introduction

- Onion roots are good samples to study cells in mitosis. As roots grow mitosis is happening in the tip of the roots. Cells in the tip of the root can be stained and viewed under the microscope.
- Stains are used to darken the components of a cell, like the nucleus, which helps distinguish cells in mitosis.
- Hematoxylin is a natural chemical extracted from a tree used to stain nuclei in microscope slide preparations.
- Methylene blue is dye that binds to components of the cells like the nucleus, DNA or RNA in the cytoplasm. It is a popular stain used for bacterial identification and structures in both animal and plant cell.

Question:

Is Hematoxylin better than Methylene Blue to stain the nucleus and see the phases of mitosis in onion root cells?

Hypothesis:

The phases of mitosis are better seen under the microscope using hematoxylin stain than Methylene Blue stain.



Investigative Methods and Procedures

Materials and supplies

- Onions
- Disposable cups
- Water
- Toothpicks
- Spatula
- Pipettes
- Safety glasses
- Lab coat
- Gloves
- Razor blades
- Tweezers
- Corning®Costar® Not Treated Multiple Well Plates
- Ruler
- Permanent markers
- Hydrochloric acid solution, HCl, 1 M
- 3% Acetic Acid with Methylene Blue
- Hematoxylin
- Cover slips
- Glass microscope slides
- Echo Revolve Microscope
- Spindrive orbital shaker platform
- Corning tubes
- Eppendorf tubes
- Absorbent pads
- Farmer's fixative solution
- Scissors

Investigative Methods and Procedures

Design and Methodology:

- My experiment was done at home and at the Center for Innovation in Brain Science (CIBS) at the University of Arizona under the guidance and supervision of the scientists Dr. Brinton, Dr. Wang and Mrs. Mao.
- First, I grew onion roots at home.
- In the lab, I cut the onion root tips and prepared them to be viewed under the microscope using two types of stains: Hematoxylin and methylene blue.
- I compared the two dyes to see which one was better at coloring the nucleus and allowing me to see the different phases of mitosis in the onion cells.

Variables:

- Independent Variable: Cellular stains
- Dependent Variable: Cell mitosis
- Control group: None used

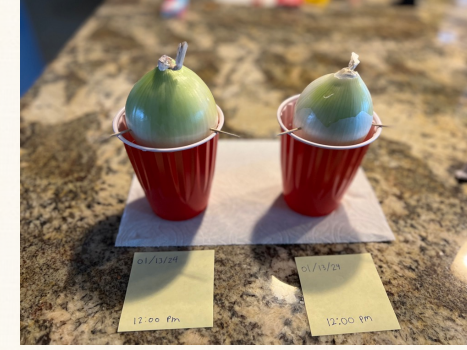
Investigative Methods and Procedures

Procedures:

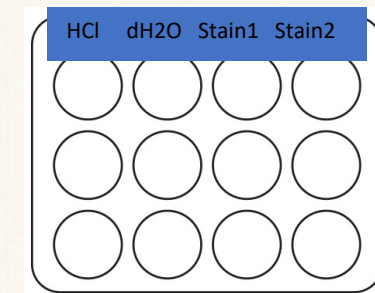
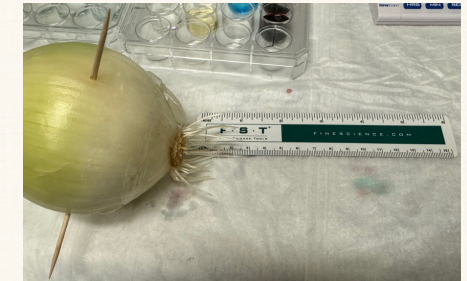
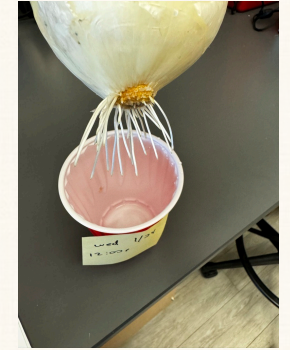
1. Fill a plastic cup with two-thirds of water. Label the cup with today's date.
2. Obtain a small onion and remove any dry outer skin.
3. Stick long toothpicks into the sides of the onion to suspend it in the water of the cup. The area with the cut roots should be submerged in the water.
4. Loosely cover the cup with plastic wrap and place it in a dark location for 72 hours.
5. After 72 hours, observe the onion roots and record your observations. If the roots are at least 2 cm in length, continue with the next step, if not place the onions back into the dark location for another 24 hours.
6. After the onion has been 2–3 days in the dark, remove it from the water and prepare the newly grown root tips for microscope analysis:
 - a. Label the well plates as follows



Petri Dishes

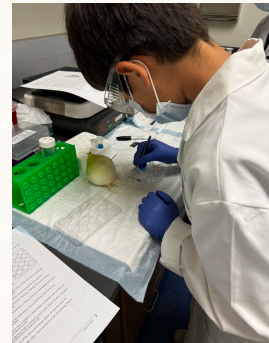
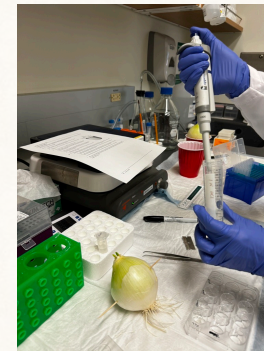
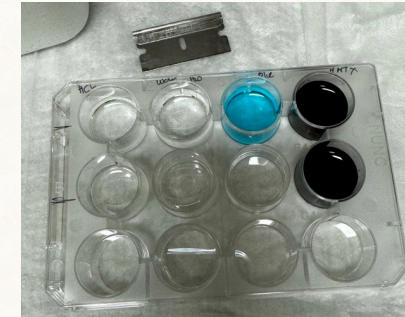
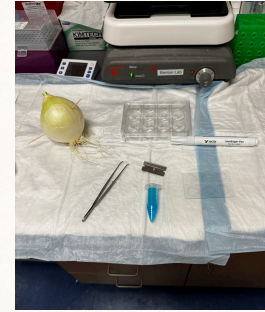


Microscope



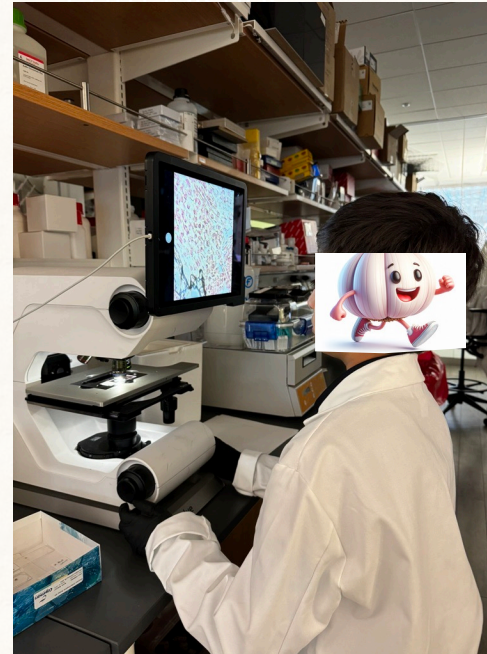
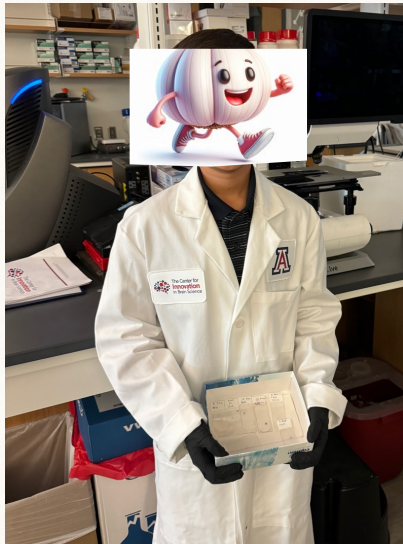
Investigative Methods and Procedures

- b. Prepare Farmer's solution: 1ml acetic acid and 3 ml ethanol.
- c. Use scissors to remove at least 4 or 5 roots from the onion. Trim the roots to approximately 1 cm in length. Discard the remainder of the root, being sure to keep the root tip.
- d. Place the roots on the Farmer's solution for at least 3 hours.
- e. Use a pipette to add to the well plates each of the solutions: 1 ml hydrochloric acid (HCl), 1ml distilled water (dH₂O), 1ml of 3 % Acetic Acid with Methylene Blue (stain 1) and 1ml Hematoxylin (Stain 2).
- f. Use tweezers to place the root tips into the HCl well for 5 minutes.
- g. After soaking the root tips for 5 minutes in the HCl, transfer them to the next well, containing dH₂O.
- h. Place the well into the Spindrive orbital shaker platform for 1 minute.
- i. Transfer the root tips to the stain (1 root stain to 3% Acetic Acid with Methylene Blue and 1 root to the Hematoxylin). Stain the tips for 5 minutes.
- j. Label the slide 1 "A red 5 min" and slide 2 "B blue 5 min". Place 1 drop of water into the slides and place the root. Cut off the root and leave 3-5 mm of the tip and place on the slide. Place a cover slip over the slide.

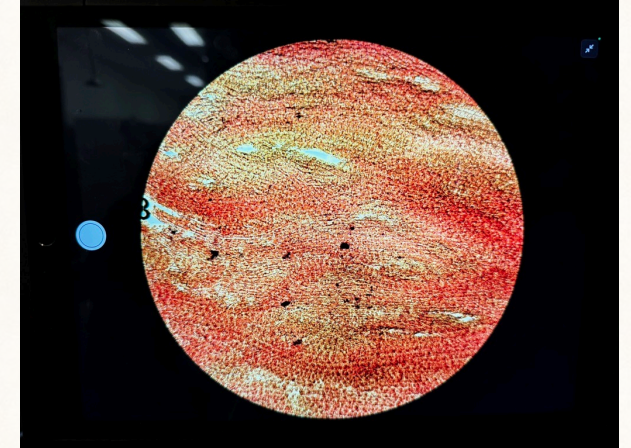


Investigative Methods and Procedures

- k. Repeat steps 6.g to 6.i. Transfer the root tips to the stain (1 root stain to 3% Acetic Acid with Methylene Blue and 1 root to the Hematoxylin). Stain the tips for 2 minutes.
 - l. Label the slide 1 "A red 2 min" and slide 2 "B blue 2 min". Place 1 drop of water into the slides and place the root. Chop the root and leave 3-5 mm of the tip into the slide. Place a cover slip over the tips.
 - m. Using the tip of a marker, press down firmly on the top slide to squash the root tips.
7. Observe the slides in the microscope.



Microscope

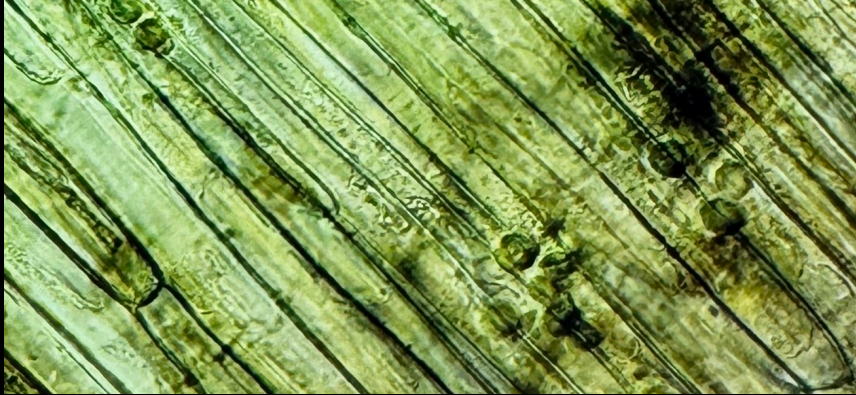


Results

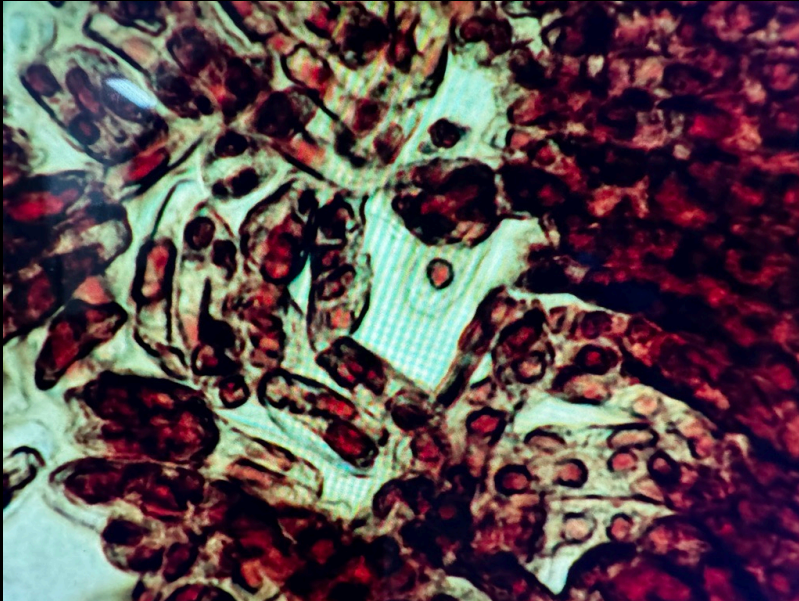
	HEMATOXYLINE	METHYLENE BLUE
Onion root cells and the nucleus color	Stained dark red	Stained light green
Mitosis clearly seen?	Yes	No
Phases of mitosis seen	Metaphase Prophase Telophase	None
Time of staining (5 min vs 2 min)	"A red 5 min" was better than "A red 2 min"	"B blue 5 min" was better than "B blue 2 min"

Results

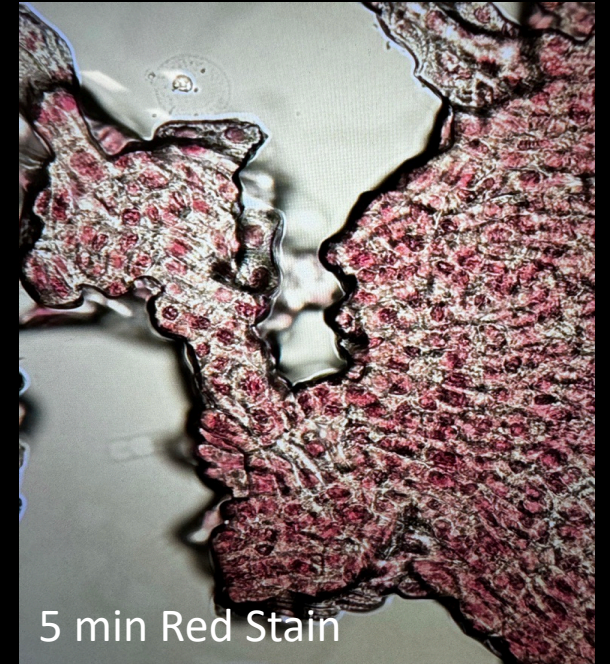
Nucleus staining with Methylene Blue



Nucleus staining with Hematoxylin (red)



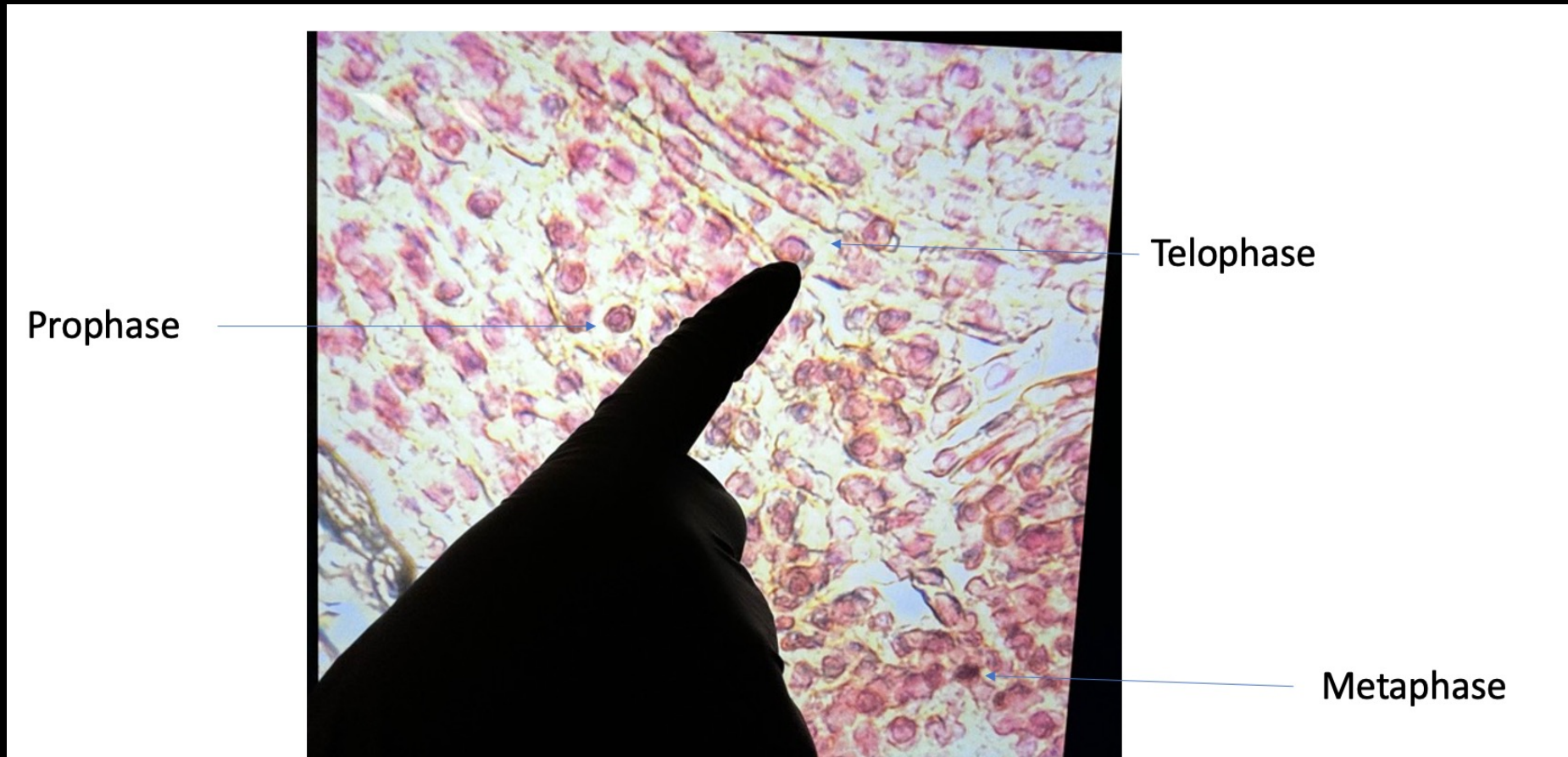
2 min Red Stain



5 min Red Stain

Results

Phases of mitosis



Discussion and Interpretation

- Onion roots grew as a result of putting the onion bulb in water which shows that cells were undergoing mitosis and replicating.
- The data show that the phases of mitosis in onion root cells are better seen under the microscope using hematoxylin stain than Methylene Blue stain.
 - The hematoxylin 5min red stain showed 3 different phases of mitosis (metaphase, prophase and telophase) and was better than the 2min red stain.
 - The longer the exposure of the root tip to the stain the better it was to visualize under the microscope.
 - The hematoxylin was better than the blue stain probably because it was able to penetrate the cell wall of the onion cells better. Cell walls are a characteristic of plant cells and not of animal cells.
- Based on the data, the hypothesis was accepted or proved correct.
- The visualization of the cells and nucleus in general might have been affected by how well the onion tips were smashed on the slide.

Implications and Ideas for Future Research

- Future experiments may rely on these data as they decide which stain to use to view the cells and nucleus of onion root tips or other plants.
- Hematoxylin can be used in a future study to determine the effects of a given compound in the mitotic index of onion root tips.
- Studying mitosis is important in medicine. Cancer is characterized by a high mitotic index that can cause the formation of tumors.

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